SQL PROJECT----- GLOBAL SUPERSTORE ANALYSIS

Dataset info:

- Dataset name=Global Superstore 2016
- Source=Kaggle
- Description=Contains sales, profit, customer, and product data for analysis.
- Table name=Superstore_data123

Questions + Work:

Question1-Find Total Sales

Query:

```
    select
        sum(
          to_number(
                  replace(
                      replace(
                          replace(sales,'.',''),
                       ',','.'),
                    '$','')
                  )
    )as total_sales
  from superstore_data123;
```

Insight:
Total sales is approximately **601 million**, which shows the overall revenue of business. Data was cleaned before calculation because the sales column had currency symbols and different number formats.

Question2- Find Total Profit

Query:

```
    select
      sum(
        to_number(
```

```
                    replace(
                        replace(
                           replace(profit,'.',''),
                     ',','.'),
'$','')
    )
) as total_profit
 from superstore_data123;
```

Insight:
Total profit is approximately **116 million**, which means the business is earning profit overall and is in a good financial position.

Question3-Find total number of orders

Query:

```
    select
       count (
           distinct order_id )
         as total_orders
    from superstore_data123;
```

Insight:
There are **15,536 total orders** in the dataset, which shows the overall number of transactions during the given period of time .

Question4- Find total quantity sold

Query:

```
    select
       sum(quantity)
           as total_quantity
    from superstore_data123;
```

Insight:
Total quantity sold is **99,816 units**, which shows total number of products sold. However, without knowing the time

period , product type, and business scale, it is difficult to say whether this is high or low.

Question5- Find total sales for each region

Query:

```
     Select
         region,
              sum(
                 to_number(
                      replace(
                          replace(
                                replace(sales,'.',''),
                             ',','.'),
                 '$','')
                )
) as total_sales
from superstore_data123
      group  by region;
```

Insight:

Sales are highest in regions such as **Southern Europe and Western Europe**, which shows strong market performance in these areas. Some regions has lower sales, which may need improvement.

Question6- Find top 5 customers by total sales

Query:

```
     select
         customer_name,
              sum(
                to_number(
                      replace(
                          replace(
                             replace(sales,'.',''),
                    ',','.'),
        '$','')
)
)as total_sales
```

```
 from superstore_data123
group by customer_name
 order by total_sales desc
fetch first 5 rows only;
```

Insight:
Ben Ferrer has the highest sales of approximately **15.2 million**, which shows that a few customers contribute a large part of total sales.

Question7- Find total profit for each category

Query:

```
   Select
        category,
          sum(
            to_number(
                replace(
                   replace(
                      replace(profit,'.',''),
            ',','.'),
     '$','')
   )
)as total_profit
from superstore_data123
group by category
 order by total_profit desc;
```

Insight:
The **Technology category has the highest profit** followed by Office Supplies while **furniture has the lowest profit.** This shows that the Technology products are more profitable as compared to other categories.

Question8- Find products where profit is negative

Query:

```
    Select
        product_name,
              sum(
                to_number(
```

```
                    replace(
                        replace(
               replace(profit,'.',''),
      ',','.'),
 '$','')
  )
) as total_profit
from superstore_data123
group by product_name
   having sum
       (to_number
            (replace
               (replace
                  (replace(profit,'.',''),
   ',','.'),
'$','')
   )
)<0
 order by total_profit;
```

Insight:

The company is overall profitable, but there are some products that are causing losses . **By focusing on these products**, the company can **improve its overall profit**.

Question9- Find total sales for each year

Query:

.....

Insight:

I tried to perform year-wise analysis using order_date, but due to incorrect date format while uploading data in oracle, the results were not accurate. So i skipped date-based analysis and focused on sales and profit analysis.

Question10- Find top 5 products by total sales

Query:

```
   select
       product_name,
```

```
          sum(sales_clean)
               as total_sales
          from superstore_data123
   group by product_name
  order by total_sales desc
     fetch first 5 rows only;
```

Insight:
Top 5 products are generating highest sales in the company. These products are very important for business and contribute major part of revenue. Company should focus more on these products to maintain and increase sales. **Company can also check why these products are performing well and apply same strategy to other products.**

Question11- Find top 5 products by profit

Query:

```
    Select
      product_name,
         sum(profit_clean)
              as total_profit
      from superstore_data123
              group by product_name
 order by total_profit desc
    fetch first 5 rows only;
```

Insight:
Top 5 products are generating highest profit for the company. These products are very important because they are contributing more to company's earnings. **Company should focus on selling more of these products.**

Question12- Find profit margin (%) for each category

Query:

```
   Select
      Category,
        sum(sales_clean) as total_sales,
```

```
  sum(profit_clean) as total_profit,
            (sum(profit_clean)/
                  sum(sales_clean))
                      *100 as profit_margin
    from superstore_data123
        group by category
            order by profit_margin
     desc;
```

Insight:

Technology category has the **highest profit margin (22%)**, so it is the most profitable category.
Furniture has the **lowest sales, profit, and profit margin (12%)**, so it is the weakest performing category and needs improvement.
Whereas **Office Supplies has highest sales** but its **profit is lower than technology**, which results in a **lower profit margin (17%)**. This means it is less efficient in generating profit.

Question13: Find average sales per order

Query:

```
    select
       sum(sales_clean)/
                count(order_id)
                   as avg_order_value
     from superstore_data123;
```

Insight:

The average order value is around 15,283, which shows that customers spend a good amount per order. This means that on average, each order generates around 15k in sales. **Increasing overall revenue without increasing the number of orders**.

Question14: Find the region which generates the highest total profit

Query:

```
select
   region,
     sum(profit_clean)
         as total_profit
            from superstore_data123
group by region
order by total_profit desc
fetch first 1 row only;
```

Insight:

Western Europe is generating the highest profit **approximately 2.25 crore** for the company. This shows that this region is performing very well. **The company can focus more on this region or invest more to maximize growth.**

Question15: Find the customer who generates the highest total profit

Query:

```
    select
      customer_name,
        sum(profit_clean)
           as total_profit
             from superstore_data123
group by customer_name
order by total_profit desc
 fetch first 1 row only;
```

Insight:

Bill Eplett contributes the highest profit approximately **6.27 lakh**, which makes them an important customer for the business. The company should **maintain a good relationship** with such customers which can **help in improving long-term profitability**.

Question16: Find which category has the highest total quantity sold

Query:

```
   Select
     category,
        sum(quantity)
           as total_quantity
             from superstore_data123
 group by category
 order by total_quantity desc
fetch first 1 row only;
```

Insight:

Office Supplies has the highest quantity which is **1,08,244** , so its **sales are high**. But its **profit is lower than Technology**, which means it earns less profit per product. **Technology is more profitable even with lower quantity.**

Challenges:

While working on this project, I faced issues with date format during data upload in Oracle. Due to this, date-based analysis was skipped.

SKILLS & LEARNINGS:

In this project, I worked on a single table dataset and performed data cleaning and analysis.

I used SQL concepts like:

- GROUP BY for grouping data
- Aggregate functions like SUM and COUNT
- HAVING CLAUSE for filtering aggregate data

I also learned how to clean data using REPLACE and TO_NUMBER functions and convert raw data into meaningful insights.

```
This project helped me understand how to analyse business
data such as sales, profit, customer, and regions.
```

Conclusion:

At the end I want to conclude that:

This project helped me understand how to use SQL for data analysis. I learned how to clean data, perform calculations, and generate insights that can help in business decision- making.